

JOHNSTON, NC, USA

WMO: 722131

Lat: 35.541N

Lon: 78.390W

Elev: 165

StdP: 14.61

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03713

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.3	24.6	-5.7	4.1	35.3	3.0	6.5	37.8	18.1	53.7	16.2	56.2	1.5	0	0.362

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	94.9	76.1	91.4	75.5	90.4	75.0	80.6	86.2	79.2	85.7	78.1	85.3	4.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	152.6	82.2	77.3	143.1	81.5	76.5	139.2	81.1	44.2	87.5	42.8	84.9	41.5	87.1	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	12.5	10.8	DB	12.9	98.3	4.5	2.6	9.7	100.2	7.0	101.8	4.5	103.2	1.2	105.2	(4)
(5)			WB	12.2	82.0	4.3	1.6	9.1	83.1	6.6	84.1	4.1	85.0	1.0	86.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.6	42.3	45.3	51.9	61.5	69.6	77.1	80.2	78.6	73.3	62.1	51.0	45.7
	DBStd	15.40	9.92	9.40	9.43	7.75	6.64	4.63	3.56	3.96	5.30	8.22	8.29	9.22
	HDD50	860	279	188	95	7	0	0	0	0	0	7	86	198
	HDD65	3053	704	554	418	156	36	0	0	0	7	153	425	600
	CDD50	5108	41	58	153	351	607	812	936	886	698	383	117	66
	CDD65	1824	1	3	11	50	179	362	471	421	255	63	5	3
	CDH74	17038	4	28	145	588	1612	3486	4721	3907	1964	526	44	13
	CDH80	6895	0	1	15	145	526	1519	2206	1697	667	115	4	0
Wind	WSAvg	3.2	3.4	4.0	4.3	4.4	3.4	2.8	2.6	2.1	3.0	2.8	3.1	2.9
Precipitation	PrecAvg	48.6	4.0	3.7	4.4	3.3	4.2	4.4	5.9	4.8	4.5	3.2	3.0	3.1
	PrecMax	68.4	8.0	6.5	7.8	8.7	9.2	14.9	14.8	10.6	18.1	15.5	6.8	7.3
	PrecMin	33.7	0.9	0.5	1.2	0.5	1.3	1.0	1.3	1.0	0.0	0.0	0.3	0.4
	PrecStd	8.0	1.7	1.8	1.8	1.7	2.1	2.7	3.1	2.3	3.2	2.7	1.7	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.2	78.6	81.7	88.5	91.5	97.0	97.5	97.2	91.0	88.1	79.1	75.5
		MCWB	64.7	64.0	61.4	63.8	71.1	75.7	75.9	76.5	74.7	73.1	65.1	67.3
	2%	DB	69.9	72.5	78.6	83.7	88.4	93.0	95.0	93.0	89.8	82.2	73.4	71.8
		MCWB	63.1	59.9	60.8	63.8	70.5	75.2	76.9	76.3	74.5	70.1	62.9	65.3
	5%	DB	64.2	68.0	73.3	81.1	85.9	90.6	91.5	90.7	87.6	80.5	70.4	67.5
		MCWB	57.7	58.5	59.2	63.9	69.9	74.2	76.9	75.7	73.0	69.0	62.6	62.4
	10%	DB	60.6	63.3	69.9	77.1	82.0	88.4	90.4	89.5	84.0	76.7	66.3	63.0
		MCWB	54.3	55.0	58.1	63.5	68.7	73.5	76.2	75.5	72.1	68.3	59.5	57.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.0	67.0	67.0	73.1	75.9	79.4	82.8	81.9	81.0	77.4	71.9	71.2
		MCDB	70.4	73.0	73.2	78.5	82.6	87.5	87.9	86.3	82.5	80.1	75.1	73.3
	2%	WB	64.0	63.9	65.1	70.2	74.1	78.1	81.0	80.2	78.2	75.0	67.5	67.1
		MCDB	67.1	68.9	71.5	76.2	81.4	87.2	87.2	85.9	82.8	79.6	70.8	70.4
	5%	WB	60.0	61.1	63.2	68.2	72.9	76.9	79.4	78.9	76.8	72.9	64.2	63.2
		MCDB	63.9	66.1	67.9	74.6	80.1	86.2	86.3	85.4	82.8	77.4	67.3	64.6
	10%	WB	54.8	56.3	60.7	66.0	71.7	75.7	78.3	77.7	75.2	70.2	61.6	58.4
		MCDB	58.3	61.7	68.1	73.8	79.1	84.3	85.9	85.0	81.4	74.1	66.2	62.2
Mean Daily Temperature Range	5% DB	MDBR	21.2	22.7	24.1	24.8	21.8	20.7	19.8	19.2	19.4	22.6	24.0	21.4
		MCDBR	27.9	28.6	29.3	28.9	24.9	23.2	21.8	21.7	23.1	24.9	27.5	25.2
		MCWBR	20.5	18.2	14.4	12.4	9.8	8.6	8.0	7.9	9.5	13.4	17.0	18.6
	5% WB	MCDBR	24.9	24.6	22.9	20.8	19.7	20.2	19.6	18.4	17.4	19.0	20.9	22.2
		MCWBR	21.4	19.1	14.9	11.4	8.7	8.2	8.5	8.1	8.4	11.2	16.0	19.4
Clear-Sky Solar Irradiance	taub	0.319	0.336	0.367	0.411	0.462	0.508	0.547	0.533	0.427	0.379	0.337	0.320	
	taud	2.479	2.425	2.374	2.250	2.146	2.063	1.969	1.999	2.293	2.403	2.479	2.513	
	Ebn at Noon	280	287	286	277	263	251	239	240	263	268	271	272	
	Edh at Noon	27	32	36	43	49	53	58	55	39	32	27	24	
All-Sky Solar Radiation	RadAvg	824	1039	1371	1731	1912	2035	1937	1742	1458	1180	906	718	
	RadStd	56	120	96	118	142	134	114	92	115	153	73	60	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+283	+203

Nomenclature: See separate page